Length-Controlled Evaluation of Training-Free Dynamic-RAG Triggers

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Abstract

Dynamic retrieval-augmented generation systems decide *when* to retrieve from scalar uncertainty signals computed at decode time. The published trigger pool (DRAGIN-RIND, FLARE, SeaKR, SelfCheck-GPT) is evaluated by raw AUROC against a binary correctness label. A converging recent literature shows that this evaluation is confounded by response length (1; 2; 3; 4; 5), but the implication has not been followed through to the dynamic-RAG trigger setting. We re-evaluate the published trigger pool under length-residualized AUROC (2) on a paired-oracle target (`lift_em`$= +1$: "does retrieval flip a wrong answer to a right one?"). Across two open-weight chat models spanning a dense vs. MoE-hybrid axis (Qwen2.5-7B-Instruct, Gemma-4-26B-A4B-it) on two multi-hop QA datasets (n=800 paired headline pool), [FACT: nine signals] retain a length-orthogonal positive margin against `tokens_generated` at nominal 95%; [FACT: four] retain it under Bonferroni correction at k=15 ($\alpha = 0.0033$). The strongest length-robust signal we identify is **ECT**, a forward-looking early-commitment-time signal at residualized AUROC [FACT: 0.691] and Δ vs. length = +0.190 [FACT: [+0.093, +0.287]] (Bonferroni lower bound +0.047); it draws on the decide-early phenomenon (6; 7; 8) and is operationally distinguished from Datta's $t^\star$ in §3.4. DRAGIN-RIND (12) also clears Bonferroni, alongside two step-level answer-trajectory aggregations and `ent.max`. We replicate the UNCERTAINTY-LINE reference implementation within [FACT: $\varepsilon \leq 0.029$ AUROC] on both architectures, and a GPT-4o-Mini semantic-equivalence judge agrees with EM labels at Cohen's κ=[FACT: 0.707] on the headline pool, under which ECT *strengthens* ($\Delta = +0.222$ vs +0.190) and 8 of 9 paired-significant survivors are preserved. Length-controlled evaluation should be standard reporting practice for dynamic-RAG triggers; the answer-trajectory primitive class is a productive direction for length-orthogonal trigger design.

1 Introduction

Dynamic retrieval-augmented generation needs triggers. Retrieval-augmented generation (RAG) systems improve factual grounding by retrieving external evidence and conditioning generation on it (28; 29). *Dynamic* RAG systems retrieve *during* generation (26; 27; 24), pausing the decode at points where the model lacks information and resuming after evidence is integrated. The control point is the **retrieval trigger**: a scalar function of the partial generation that decides *when* retrieval is invoked. Triggering too often inflates latency; triggering too late propagates hallucinations through multi-step reasoning (12; 13).

This paper focuses on *training-free* triggers, which operate at decode time without per-architecture fine-tuning. The published pool spans four signal families: DRAGIN (12) fires on a composite entropy and attention score; FLARE (13) uses token probability look-ahead; SeaKR (14) uses a Gram determinant of K=20 stochastic hidden states; SelfCheckGPT (15) uses K-sample text consistency. Trained alternatives such as Self-RAG (16), MBA-RAG (Tang et al.), CtrlA (18), DTR (19), and DioR (20) require per-architecture supervised data, and we set them aside in this paper to keep the comparison within the training-free sub-family.

All published triggers are evaluated by the same protocol. Compute the trigger score per trace, rank, and report raw AUROC against a binary correctness label (typically exact-match equality, EM = 0, on the closed-book answer). The signals above were each introduced and evaluated under this protocol.

Recent work shows the evaluation protocol is length-confounded. A converging recent literature identifies response length as a dominant confound in language-model uncertainty quantification. (1) prove that when both the uncertainty score and the correctness function share a length confounder, the estimated AUROC is *systematically skewed*, and rankings flip across correctness functions. Empirically, token-length Spearman ρ reaches -0.7 with Perplexity and -0.7 with Semantic Entropy. (3) demonstrate that "simple heuristics based on response length can rival complex hallucination detection techniques," with Perplexity dropping [FACT: 45.9%] and EigenScore [FACT: 30.4%] when the correctness function is switched. (4) document the behavioral mechanism: LLMs exhibit *verbosity compensation* (verbose responses when uncertain; GPT-4 frequency [FACT: 50.4%]; performance gap [FACT: 27.6%] on Qasper) across HotpotQA, MuSiQue,

and 2WikiMHQA, the same multi-hop QA family used here. (5) additionally show that approximate correctness functions disagree substantially on UQ rankings, with length as one driver. The implication for the trigger-evaluation protocol is direct: if trigger scores correlate with response length and length correlates with correctness, then raw AUROC rewards signals that are partial proxies for length, regardless of whether they carry any deeper signal about retrieval need.

Uncertainty-Line: regression-residualization as the fix. (2) show that nominally length-normalized UQ measures (Perplexity, Mean Token Entropy, Token-SAR) retain residual length bias, and propose UNCERTAINTY-LINE, a regression-residualization protocol: $\hat{u}_{\text{LINE}}(\hat{y}, x) = \hat{u}(\hat{y}, x) - \hat{f}(L(\hat{y}))$. The residual is used as the length-invariant uncertainty estimate; AUROC computed on the residual then asks whether the signal carries information *beyond* length. The protocol is post-hoc, model-agnostic, and applicable to any scalar trigger score. (2) evaluate it on machine translation, summarization, and mathematical reasoning; to our knowledge it has not been applied to the dynamic-RAG trigger pool.

This paper. We re-evaluate the published training-free dynamic-RAG trigger pool under length-residualized AUROC (2) on multi-hop QA with paired oracle re-decode. The published triggers were each proposed before the length-confound literature converged; we ask which of them carry a length-orthogonal positive margin and which were implicitly tracking length all along.

A length-orthogonal trigger class is emerging in concurrent work. A separate, concurrent literature studies step-to-step answer dynamics during chain-of-thought reasoning. (6) (ICML 2026) define *forced answer completion*: at each partial reasoning prefix, append a closing token and sample an intermediate answer A_i, yielding an answer trajectory $T = (A_0, \ldots, A_n)$. They define $\text{AnswerSwitches}(T) = \sum_i \mathbb{1}[A_i \neq A_{i+1}]$ and the final-switch index $t^\star = \max\{i : A_i \neq A_{i+1}\}$, and find that 68% of Qwen3-4B traces never change answer, while the model generates $\sim$760 tokens after the final switch. They apply this primitive to *early stopping* (saving compute), not to retrieval triggering. (7) (NAACL 2025) independently use a *Chain-of-Probe* that reads a closed-answer-set softmax after each step and report that traces that change answer mid-reasoning have $> 20\%$ lower accuracy than early-commit traces. (8) (Feb 2026) define a per-token instability signal

$I_t = \mathrm{JSD}(\tilde{p}_t, \tilde{p}_{t-1}) + \lambda H_t$ and report that the *timing* of the peak matters: early peaks correlate with 46% accuracy (corrective) and late peaks with 14% (destructive), a $> 3\times$ gap conditional on timing alone.

This class of signals operates on answer-set semantics rather than on token-probability or attention magnitude. As we show empirically in §4.1 and confirm via Spearman ρ in Appendix B, these signals carry markedly lower length correlation than the published trigger pool, which makes them candidate length-orthogonal triggers. We test this class under length-controlled evaluation on retrieval triggering, the first such application of the primitive to our knowledge. The strongest signal we identify is **ECT** (early-commitment time, defined in §3.4), a forward-looking commitment-time signal for non-thinking-mode chat models, drawing on the decide-early phenomenon (6; 7; 8).

Terminological note. (9) (ICML 2026, ARS) use "answer stability" to denote robustness of a decoded answer to small *latent perturbations* at the trace boundary, achieving 86.6% AUROC on TruthfulQA via supervised contrastive representation shaping. Our use of "step-level answer-trajectory triggers" denotes step-to-step disagreement under K-sample boundary-completion *sampling*: a training-free, decode-time signal, categorically different from the latent-perturbation-based ARS family.

A paired-oracle target for the deployment question. Standard trigger evaluations use the binary correctness label `closed_em`=0 (closed-book wrong) as the target. This conflates two failure modes that a deployment system handles differently: traces where retrieval would have flipped a wrong answer to a right one, and traces where the model would have been wrong regardless. Following the question raised by (30), we use a paired-oracle target

$$\texttt{lift_em}(q) = \mathbb{1}\Big[\mathrm{EM}(a_o^{(q)}) > \mathrm{EM}(a_c^{(q)})\Big]$$

that separates the two cases by re-decoding each closed-book trace under oracle context and labelling positive only when retrieval helps. The paired design shares the same exact-match correctness function across the closed and oracle arms, partially cancelling the correctness-function-bias critique of Ielanskyi et al. (5).

Contributions.

1. **Length-controlled trigger evaluation for dynamic RAG.** Under length-residualized AUROC (2) paired with the deployment-relevant

`lift_em`=+1 target above, we re-evaluate the published trigger pool on a n=800 paired headline cohort. We replicate the reference implementation (`externalCodeAudits/uncertainty-line/`, our `analysis/uncertainty_line_audit.py`) within [FACT: $\varepsilon_{\text{protocol}} \leq 0.029$ AUROC max] on both architectures ([FACT: Qwen 0.029, Gemma 0.027]).

2. **Cross-architecture re-evaluation.** We re-evaluate [FACT: 16 published trigger signals] on two open-weight chat models spanning a dense vs. MoE-hybrid axis (Qwen2.5-7B-Instruct, 7.6B dense; Gemma-4-26B-A4B-it, ∼3.8B active / 25.2B total, MoE with sliding-window/global hybrid attention) on two multi-hop QA datasets (HotpotQA-distractor in-domain, 2WikiMultiHopQA-distractor out-of-domain), at n=200 paired traces per cell (n=800 paired headline pool).

3. **Length-robust signals.** [FACT: Nine signals] retain a length-orthogonal positive margin against `tokens_generated` at nominal 95% paired-bootstrap, and [FACT: four signals] retain it under Bonferroni correction at k=15 ($\alpha = 0.0033$). The Bonferroni-robust set spans four signal families: the attention-based DRAGIN-RIND (12); the logit-family `ent.max`; the step-level answer-trajectory aggregation `min_majority`; and **ECT** (early-commitment time, defined in §3.4), the strongest single signal in our pool, at residualized AUROC [FACT: 0.691] and Δ vs. length [FACT: = +0.190 [+0.093, +0.287]] (Bonferroni lower bound [FACT: +0.047]). ECT draws on the decide-early phenomenon (6; 7; 8); operational differences from Datta's $t^\star$ are tabulated in §3.4.

4. **Multi-sample ablation.** We ablate the K=5 multi-sample variant of `flip_rate` against the K=1 Datta-style single-sample proxy and find [FACT: equality at our power] on both architectures. The multi-sample variant has no significant advantage for retrieval triggering; we recommend K=1 for deployment compute efficiency.

5. **Correctness-function robustness.** As a direct response to the Janiak et al. (3), Santilli et al. (1), and Ielanskyi et al. (5) critiques of EM- / lexical-overlap-based evaluation, we re-label both arms of every paired trace via a GPT-4o-Mini semantic-equivalence judge (Janiak's exact choice) and re-run the full survivor matrix. EM and the judge agree at Cohen's κ=[FACT: 0.707] on the headline pool. Eight of the nine paired-significant survivors under EM remain paired-significant under the judge; the headline signal ECT *strengthens* under the judge

($\Delta = +0.222$ [+0.132, +0.312] vs +0.190 [+0.093, +0.287] under EM). See Appendix C.

Scope. The answer-trajectory primitive class belongs to the field (6, 7, 8); we credit it explicitly and disambiguate ECT from $t^\star$ in §3.4. This paper is a length-controlled re-evaluation of the published trigger pool and a positioning of the answer-trajectory class as length-orthogonal trigger candidates; it is not a deployment study, and our claims pertain to signal-level discriminative power rather than end-to-end pipeline metrics (see §6).

2 Related Work

2.1 Dynamic-RAG Triggering Methods

Dynamic-RAG triggers can be grouped by the signal class they exploit.

Token-probability triggers. FLARE (13) fires when any token in a look-ahead sentence has $p < \theta$ under the model's own predictive distribution. We include `flare.max` $= 1 - \min(p)$ over each step boundary as the natural signal-level analogue.

Attention-conditioned triggers. DRAGIN (12) defines a real-time information-need score (RIND) combining token entropy, self-attention, and semantic features; we audit `dragin_rind_max.max` (the maximum over step boundaries of the DRAGIN-RIND score). DR-RAG (22) and Layered Query Retrieval (23) extend the attention-based family.

Multi-sample text-consistency triggers. SelfCheckGPT (15) measures sentence-level NLI prob(contradict) or n-gram surprisal across K stochastic completions. We include `u_cons.max` (a K=5 text-consistency proxy) and `sc_ngram` (n-gram variant). SAC^3 (35) extends the consistency family with semantic-aware sampling. Self-Consistency (34) established multi-sample answer voting as a primitive.

Hidden-state geometric triggers. INSIDE (31) introduces the Eigen-Score on hidden-state covariance matrices; SeaKR (14) extends INSIDE to triggering via a K=20 Gram determinant. We include `seakr_k5_proxy.logdet` (the K=5 proxy at our sample budget). Other hidden-state CRV signals we include are `drift_cosine.max`, `variance_l2_window.max`, and `hidden_norm.range`. CtrlA (18) extracts trained honesty features from hidden states.

Trained / fine-tuned alternatives (excluded by scope). Self-RAG (16) learns reflection tokens. MBA-RAG (Tang et al.) learns a multi-armed-bandit retrieval policy. DTR (19) learns an uncertainty-guided triggering policy. DioR (20) learns a cognitive-detection retrieval policy. DyPRAG

(21) performs parametric retrieval-knowledge injection at test time. These are training-required methods orthogonal to our training-free audit scope.

Adaptive complexity classifiers. Adaptive-RAG (25) routes by predicted query complexity. CRAG (24) predicts retrieval quality as Correct/Ambiguous/Incorrect. Both are conceptual parents of our paired-oracle target but use a different operationalization (classifier outputs vs. paired EM differences).

The parent question of paired-oracle evaluation. (30) first formalized the question *"for which questions does retrieval help?"* via popularity-conditioned analyses. Our `lift_em` target is a direct operationalization of this question on the trigger-evaluation side.

2.2 Length as a Confound in UQ Evaluation

(1) prove formally that when both the uncertainty score $\hat{u}$ and a noisy correctness function $\tilde{Q}$ depend on a shared confounder (response length L), the AUROC estimator is systematically skewed, not merely noisy. Their empirical Spearman correlations (Table 2 of their paper) show $\rho(\text{length}, \text{Perplexity}) = -0.7$, $\rho(\text{length}, \text{Naive Entropy}) = -0.6$, $\rho(\text{length}, \text{Semantic Entropy}) = -0.7$, and Token-Length itself is a competitive UQ-method baseline on their leaderboard.

(2) propose Uncertainty-Line: regress the UQ score on length and use the residual as the length-invariant estimate. They report consistent improvements across NMT, summarization, and mathematical reasoning. Public code: `https://github.com/stat-ml/uncertainty-line`.

(3) (EMNLP 2025) report that "*simple heuristics based on response length can rival complex hallucination detection techniques.*" Switching ROUGE to LLM-as-Judge labels drops Perplexity AUROC by [FACT: 45.9%] and EigenScore by [FACT: 30.4%] on NQ-Open, SQuAD, and TriviaQA.

(4) identify *Verbosity Compensation* (VC) as the behavioral mechanism: LLMs systematically generate longer, wordier responses when uncertain. GPT-4 VC frequency is [FACT: 50.4%]; verbose-vs-concise performance gap is [FACT: 27.6%] on Qasper. The verbosity–uncertainty link holds across all five evaluated datasets including HotpotQA, MuSiQue, and 2WikiMHQA (the same multi-hop QA family used in this paper).

(5) show that approximate correctness functions disagree substantially on UQ method rankings. Pairwise mutual AUROC drops as low as [FACT: 0.53], and Spearman ρ between rankings is sometimes *negative* ($\rho = -0.66$ for BLEU vs. ROUGE on CoQA). They identify length as one driver and recommend marginalizing over correctness functions, structured tasks, and

Elo aggregation.

Acknowledged-gap citations. (11) state that "*joint generation probability suffers from severe length bias*," but propose no residualization fix. (8) cap output to 128 tokens as a length control but do not evaluate under regression-based residualization.

The gap we cross. The five foundation papers establish the length confound on general UQ benchmarks. None audits the specific dynamic-RAG-trigger pool, which is the trigger set most directly affected by length-correlated correctness on multi-hop QA.

2.3 Step-Level Answer-Trajectory Primitives

(6) (ICML 2026) is the direct primitive parent. Their *forced answer completion* probes intermediate predictions after each reasoning step on a thinking-mode reasoning model (Qwen3-4B/8B/30B-A3B), defines AnswerSwitches$(T) = \sum_i \mathbb{1}[A_i \neq A_{i+1}]$ and $t^\star = \max\{i : A_i \neq A_{i+1}\}$, and applies these to *early stopping*. Probe-based stopping at the detected final switch reduces reasoning tokens by $\sim$500 per query at $<$2% accuracy drop. Our `flip_rate`, `late_flips`, and ECT signals adapt their primitive to retrieval triggering on non-thinking models.

(7) (NAACL 2025) introduce *Chain-of-Probe*, the closest soft-confidence analogue. They append a fixed probing string after each step and read the softmax over a closed answer set. Their finding that "*changing-during-reasoning has >20% lower accuracy than early-answering*" has the same direction of effect as our `flip_rate` vs. correctness pattern.

(8) introduce the timing distinction. The peak instability position $\rho = t^\star/T$ discriminates corrective (early, recoverable) from destructive (late, irrecoverable) instability with a $>3\times$ accuracy gap. Their token-distribution-level JSD signal I_t is the parameter-free probing cousin of our answer-level `late_flips` aggregation.

(9) (ICML 2026, ARS) develop a *supervised representation-shaping* approach to "answer stability" via latent perturbations and contrastive training, reaching 86.6% AUROC on TruthfulQA. ARS sits in a different family (supervised representation training, not training-free decode-time sampling). We cite it to disambiguate the term "answer stability," which we avoid in favor of "step-level answer-trajectory triggers."

2.4 Stepwise and Streaming Uncertainty in Chain-of-Thought

(10) (the trained-probe upper bound on the same Qwen2.5-7B backbone we

audit) train step-level + prefix-level probes on hidden states with Claude-4.5 labels, achieving [FACT: 87–93% AUROC]. (11) (ACL 2026) mine discriminative Signal Temporal Logic formulas over stepwise confidence trajectories, achieving AUROC ∼0.62 parameter-free. (6) additionally provide a hold-for-k causal smoothing operator (Eq. 4) that admits streaming operation. Our work is in the parameter-free, training-free sub-family of this temporal-modeling research front, with the specific application of retrieval triggering.

2.5 Hidden-State Probes (Out of Scope, Cited for Completeness)

(10), (9), (18), and (36) train hidden-state probes on labelled data. (37) learn unsupervised truthfulness directions in representation space. These methods can in principle adjoin our paired-oracle protocol, but they require per-architecture labelled data and training infrastructure. We exclude them from the training-free audit and treat them as future work.

3 Methods

3.1 Paired-Oracle Protocol and the `lift_em` Target

For each question q in the evaluation set, we run two decodes:

1. **Closed-book** $(\tau_c^{(q)}, a_c^{(q)})$: greedy generation with no retrieval context, producing the answer $a_c^{(q)}$ and the *decode trace* $\tau_c^{(q)}$ from which we extract all trigger signals.

2. **Oracle-retrieval** $(\tau_o^{(q)}, a_o^{(q)})$: greedy generation conditioned on the gold supporting paragraphs (the *oracle context*; for HotpotQA the `supporting_facts` sentences, for 2WikiMHQA the `evidences` field). All trigger signals are extracted from $\tau_c^{(q)}$ only; the oracle decode is used solely to label the trace.

The discrimination target is

$$\texttt{lift_em}(q) = \mathbb{1}\Big[\mathrm{EM}(a_o^{(q)}) > \mathrm{EM}(a_c^{(q)})\Big] \in \{0, 1\}. \tag{1}$$

A positive label denotes "retrieval would have flipped this trace from wrong to right." This formalizes the question raised by (30) on which traces a deployed dynamic-RAG system should trigger retrieval for.

Why paired-oracle. A standard alternative target, `closed_em` $= 0$ (closed-book wrong), conflates two sources of error: (a) the model could have answered correctly with retrieval; (b) the model will be wrong regardless. Triggering retrieval on (b) wastes compute and may introduce noise. The `lift_em`$= +1$ target isolates (a).

Defense against correctness-function bias. (5) show that approximate correctness functions can disagree substantially across UQ-method rankings, with pairwise mutual AUROC dropping as low as 0.53 for some pairs on the COQA benchmark. (3) report that switching ROUGE-L $\rightarrow$ LLM-as-Judge labels drops Perplexity AUROC by 45.9% and EigenScore AUROC by 30.4%. Our paired design in Eq. (1) is partially insulated by construction: both arms of `lift_em` share the same exact-match correctness function, so EM-specific bias and length-correlated lexical-overlap bias partially cancel within each paired comparison. This is a structural defense but not a substitute for a direct test.

LLM-as-Judge robustness check. To test the survivor matrix under a less length-biased correctness function, we re-label both arms of every paired trace via a `gpt-4o-mini` semantic-equivalence judge (3's choice). The judge issues a binary YES/NO verdict under a strict prompt, and we recompute the paired-oracle target as $\texttt{lift_em}_{\text{JUDGE}}(q) = \mathbb{1}[\text{judge}(a_o^{(q)})\text{=YES} \wedge \text{judge}(a_c^{(q)})\text{=NO}]$, then re-run the full survivor-matrix analysis with paired bootstrap B=10,000. EM and judge labels agree at Cohen's $\kappa =$ [FACT: 0.707] on the headline pool — substantial agreement per the Landis & Koch (41) scale. Eight of the nine paired-significant survivors under EM remain paired-significant under the judge, with one out-in swap that does not affect the Bonferroni-robust four. Full results in Appendix C.

Code. `analysis/r1_phi_paired_audit.py` (per-arch paired-audit driver; applied to Qwen and Gemma cells), `analysis/r1_phi_d1_closed_audit.py` (helper module providing the shared signal extractors, AUROC, OLS-residual, and paired-bootstrap primitives); trace generation via `scripts/04f_run_qa_sota.py` (closed) and `scripts/04l_oracle_retrieve_decode.py` (oracle).

3.2 Length Residualization

For each trigger signal s on a cohort of paired traces, we compute

$$s_{\text{resid}} = s - \hat{\alpha} - \hat{\beta}\,\log(\texttt{tokens} + 1), \tag{2}$$

where $\hat{\alpha}, \hat{\beta}$ are ordinary-least-squares coefficients fit on the cohort. AUROC is then evaluated on s_{resid} against the target label. We use $\log(\texttt{tokens} + 1)$ rather than raw length to mitigate overcorrection on the long-tailed token distribution in multi-step reasoning traces, consistent with the caveat in (2, §3).

Replication of Uncertainty-Line. To pre-empt the reimplementation critique, we run a head-to-head replication of the reference UNCERTAINTY-LINE implementation (2; `externalCodeAudits/uncertainty-line/utils.py::detrend_ue`). The reference math is MinMax-scale signal and length, fit `LinearRegression(signal_norm ~ length_norm)` on a 70/30 train/test split with 5%–95% length trim, then residualize on the test set. We report (i) our protocol on the full cohort (Eq. 2); (ii) their math on the full cohort (apples-to-apples, $\varepsilon_{\text{protocol}}$); (iii) their math with their train/test split, averaged over 5 seeds (sampling-noise-isolated, $\varepsilon_{\text{sampling}}$).
Results: $\varepsilon_{\text{protocol}}$ max is [FACT: 0.029 (Qwen)] and [FACT: 0.027 (Gemma)], both well below the typical reviewer-attack threshold of 0.05. The $\varepsilon_{\text{sampling}}$ component reflects sampling noise at n=200, not a protocol difference. See `analysis/uncertainty_line_audit.py` and `analysis/uncertainty_line_audit_*.out,json`.

Statistical significance. Paired bootstrap of Δ(signal AUROC$-$tokens_generated AUROC) with B=10,000 resamples, 95% percentile confidence intervals. Bonferroni correction across k=15 signals ($\alpha = 0.0033$, ∼99.67% CI) reported in Appendix A as the conservative cross-check; main-body claims are at nominal $\alpha = 0.05$.

Code. `analysis/r1_phi_paired_audit.py` (R1 paired-audit driver, used for both Qwen and Gemma cells); `analysis/moonshot_explore.py` + `analysis/moonshot_ensemble.p` (the headline survivor-matrix and ensemble drivers); `analysis/_common.py` (AUROC, paired bootstrap, OLS residual helpers).

3.3 Trigger-Signal Pool

We evaluate 16 baseline signals spanning the four major families surveyed in §2.1 plus length baselines, summarized in Table 1. All signals are extracted from the closed-book trace τ_c and aggregated to a trace-level scalar via the indicated reduction (typically max over step boundaries).

Scope. We restrict to *training-free* signals. Self-RAG (16), MBA-RAG (Tang et al.), CtrlA (18), DTR (19), DioR (20), and ARS (9) are out of scope by design; they require per-architecture supervised data.

Implementation note: aggregation choice. All signals reduce to a trace-level scalar (typically max over step boundaries). This convention matches the original DRAGIN/FLARE/SeaKR threshold-firing semantics with maximum-aggregated step score. We do *not* replicate the full deployment pipelines (threshold tuning, retriever choice, prompt format). Our claims pertain to signal-level discriminative power on the `lift_em` $= +1$ target, not method-level end-to-end performance (see §6).

3.4 Step-Level Answer-Trajectory Triggers

At each reasoning-step boundary (detected by spaCy sentence segmentation with 32-token fallback), we sample K=5 boundary completions at temperature τ=0.6, top-p 0.9, with answer extraction via the regex `r"\bAnswer\s*:\s*(.+?)"`. We aggregate these boundary samples into four *step-level answer-trajectory triggers* operating on the `AnswerSwitches`-style primitive class (6):

- `flip_rate K=5`: at each step, take the majority answer over K=5 samples; count consecutive-step majority disagreements, normalize by the number of answer-bearing steps.
- `flip_rate K=1`: same but with the first sample only (K=1, drawn from the same τ=0.6, top-p 0.9 sampler). We use this as a directional proxy for Datta's deterministic-greedy forced-completion; the proxy is not byte-faithful (Datta uses greedy decoding after appending the answer prefix, whereas our `samples[0]` is the first stochastic completion under the configured K=5 sampler). A byte-faithful Datta replication is future work (§7); the proxy is good enough for the directional regime-2 conclusion below.
- `late_flips`: count only flips occurring in the last 50% of the trace. Motivated by the corrective/destructive timing distinction of (8): late flips correlate with destructive (irrecoverable) instability and higher retrieval need.
- `min_majority`: $1 - \min_t(\max_{count}/K)$, the maximum *within-step* disagreement across the trace. Captures sample disagreement at the

worst single step rather than between-step transitions. This is a K-sample analogue of the SelfCheckGPT (15) family at the answer level rather than the sentence level.

We do not coin a family name for these aggregations; the primitive class is the `AnswerSwitches` trajectory of Datta et al. (6), and we cite that work as the parent throughout. The four aggregations above are operational variants applied to retrieval triggering on non-thinking-mode chat models.

K=5 vs. K=1 ablation. A natural question is whether the multi-sample K=5 aggregation has any empirical advantage over the single-sample K=1 Datta original. We ablate this across both audited architectures (§4.3) and find *regime 2 equality* on the `lift_em` target: the K value is empirically free at our power. We recommend K=1 for deployment compute efficiency.

ECT: Early-Commitment Time. Beyond the four aggregations above, we propose an additional signal motivated by the "decide-early" phenomenon documented by Datta et al. (6), Wang et al. (7), and Chen et al. (8).

Definition (ECT). Let $\mathcal{M} = (m_1, m_2, \ldots, m_n)$ be the sequence of per-step K=5 majority answers (skipping non-answer-bearing steps), and let a_{final} be the final extracted answer of the trace. Define

$$t_{\text{commit}} = \min\{i \ : \ m_j = a_{\text{final}} \text{ for all } j \geq i\}, \tag{3}$$

i.e., the earliest step from which the majority remains stable and equal to the final answer for the remainder of the trace. The scalar trigger score is $\text{ECT} = t_{\text{commit}} / \max(1, n-1) \in [0, 1]$. Late commitment (high score) corresponds to extended deliberation. On retrieval-helpful traces the model has not yet converged in the closed-book decode, and external evidence resolves the answer.

Relationship to Datta's $t^\star$. ECT is motivated by the final-switch index $t^\star = \max\{i : A_i \neq A_{i+1}\}$ of Datta et al. (6), but it is operationally distinct. Table 2 lists the three operational differences.

The *decide-early phenomenon* belongs to the field (6; 7; 8). The specific t_{commit} formulation in Eq. (3) and its application to dynamic-RAG triggering are ours.

3.5 Architecture Pool

We evaluate the headline trigger-survivor matrix on two open-weight chat models spanning two architecture families: a dense Transformer (Qwen2.5-7B-Instruct) and a Mixture-of-Experts model with hybrid sliding-window/global attention (Gemma-4-26B-A4B-it, 3.8B active / 25.2B total). We chose this pair to span the dense-vs-MoE-hybrid axis at the open-weight 7B+ tier.

Trace capture schema (rq1-pilot-v1). For each step boundary, we log: `mean_token_entropy`, `boundary_token_entropy`, `per_token_max_logprob` (post and pre repetition penalty), `dragin_rind_max`, `drift_cosine`, `variance_l2_window`, `hidden_norm`, `hvr` (SELFDOUBT-T1), K=5 `samples` (text + EOS hidden state + finish reason), and per-trace metadata (`tokens_generated`, `wall_clock_s`, `em`, `f1`, gold answer, question type, etc.). Schema docs in `scripts/crv/sota_capture.py`.

3.6 Datasets

We use two multi-hop QA distractor benchmarks:

- **HotpotQA-distractor** (43) as the in-domain multi-hop QA benchmark. Both bridge and comparison question types are included; the test split is drawn from the original validation set (`n`=500 per arch, first 200 used here).
- **2WikiMultiHopQA-distractor** (44) as the out-of-domain benchmark. Question types include bridge, comparison, compositional, inference, and bridge_comparison.

Both datasets are widely used in the dynamic-RAG-trigger literature (12; 14; Tang et al.; 4), so our results are directly comparable.

Sample slice. For each architecture, we use a fixed 500-question test slice; we report headline numbers on the first 200 questions of each slice (`n`=200/cell, 4 cells per arch). Slice manifest at `runs/v2_headline_`*`arch`*`/checkpoints/slice_manifes`

MuSiQue. (45) provides a third multi-hop QA benchmark with longer reasoning chains; we discuss its inclusion as future work (§7) but do not include it in the current audit due to compute constraints.

3.7 Statistical Analysis

Every numerical claim in this paper carries a paired-bootstrap (B=10,000) 95% confidence interval on Δ = AUROC(s_{resid}) − AUROC(tokens), where s_{resid} is the OLS residual of the signal on log(tokens + 1) on the same cohort. A signal is declared a *length-robust survivor* if the lower bound of the CI exceeds zero. Per-architecture results are reported at the cell level (arch × dataset, n=200) and pooled (arch × datasets, n=400). The headline pool aggregates Qwen+Gemma × HotpotQA+2Wiki at n=800.

Multiple-testing correction. We additionally report Bonferroni correction at k=15 ($\alpha = 0.0033$, ∼99.67% CI) as a robustness check on the survivor set (Appendix A). **Four signals** (ECT, DRAGIN-RIND, `min_majority`, `ent.max`) clear paired-significance under Bonferroni; the remaining five nominal-95% survivors fall back to the borderline band under correction.

Per-question-type stratification. We additionally report per-type AUROC on both datasets: HotpotQA {bridge, comparison}; 2WikiMHQA {bridge_comparison, comparison, compositional, inference}.

4 Results

4.1 Cross-Architecture Survivor Matrix

Figure 1 visualizes the headline pool (Qwen + Gemma, n=800 paired, [FACT: 108 helped] traces, `lift_em`= + 1 base rate ≈ [FACT: 0.135]) as a forest plot of Δ = AUROC(s_{resid}) − AUROC(`tokens`) with paired-bootstrap 95% and Bonferroni-corrected ∼99.67% CIs ($\alpha = 0.05/15 = 0.0033$, k=15, B=10,000). Three tiers emerge: four signals are Bonferroni-robust; five more clear at nominal 95% only; the remaining eight (mostly hidden-state) hug the length baseline. Crucially, the four Bonferroni-robust signals span four different signal families (attention, logit, K-sample answer-trajectory aggregation, and the answer-trajectory commitment-time **ECT**). Table 4 reports the same matrix with exact numbers.

Table 4 presents the same matrix in tabular form. All values are length-residualized AUROC; the rightmost columns report Δ vs. `tokens_generated` with paired-bootstrap (B=10,000) at nominal 95% CI and at Bonferroni-corrected ∼99.67% CI.

4.2 The Strongest Length-Robust Survivor: ECT

The strongest signal under length residualization on the headline pool is **ECT**, our forward-looking early-commitment-time signal (Eq. **??**, §3.4). It is defined as the earliest step from which the K=5 majority answer remains stable and equal to the final answer for the rest of the trace. ECT draws on the "decide-early" phenomenon established by Datta et al. (6), Wang et al. (7), and Chen et al. (8), but it is operationally distinct from Datta's last-switch index $t^\star$ on three axes (temporal direction, sampling regime, application target; Table 2).

Per-cell ECT raw AUROC. Table 5 shows per-cell raw AUROC of ECT on `lift_em` = +1:

Why ECT works. The early-commitment-time signal exploits the observation from Datta et al. (6) and Wang et al. (7) that models decide early on questions they will get right and continue to deliberate on questions where retrieval would help. The retrieval-helpful subpopulation is the same subpopulation where the answer trajectory has not stabilized. ECT captures this stabilization delay in its forward-looking formulation. A low ECT score means the model committed early (closed-book usually right); a high ECT score means deliberation persists through the trace (retrieval often helps). The use of K=5 boundary-completion majority as the answer operator allows the signal to be computed on non-thinking-mode chat models, where the explicit forced-completion protocol of Datta et al. (6) (which requires a `</think>` token plus task-specific answer prefix) is unavailable.

Attribution. The "decide-early" phenomenon is established by Datta et al. (6), Wang et al. (7), and Chen et al. (8) (each with a different operationalization: last-switch $t^\star$, closed-set softmax probing, and JSD timing-peak respectively). The ECT formulation in Eq. (**??**) is ours, as is its application to dynamic-RAG triggering and its characterization as the strongest length-robust survivor in the audited trigger pool. We do not claim the answer-trajectory phenomenon as a novel observation. We claim the specific forward-looking K=5-majority operationalization and the retrieval-triggering application.

4.3 K=5 vs. K=1: The Multi-Sample Ablation is Null

A natural question is whether the K=5 stochastic-sample variants of the answer-trajectory triggers have empirical advantage over the K=1 single-sample Datta original. Table 6 reports the $\Delta(K{=}5{-}K{=}1)$ on the `flip_rate` aggregation across both audited architectures.

Regime-2 verdict. Across both audited architectures (dense Transformer and MoE-hybrid), the K=5 multi-sample variant of `flip_rate` shows no significant advantage over the K=1 Datta original on the retrieval-triggering target. We pre-registered three outcome regimes for this ablation before observing the cross-architecture data: (1) K=5 paired-sig wins; (2) $K{=}5 \approx K{=}1$, CIs straddle zero; (3) K=1 paired-sig wins. Regime 2 is the observed outcome on both architectures on the headline target. We **recommend** K=1 for deployment for compute-efficiency reasons. A single-sample boundary-completion probe is 5× cheaper than the K=5 alternative with equivalent triggering performance at our power.

Honest scope of the claim. On the secondary `closed_em`=0 target, Gemma shows K=5 marginally above K=1 (raw $\Delta = +0.063$ [+0.016, +0.112], paired-sig at the lower edge). This is a K-sample advantage on the incorrectness target, not on the retrieval-helpfulness target. The pattern is consistent with K-sample disagreement being a generic uncertainty signal that does not necessarily encode the oracle-helpfulness substructure. We discuss this in §5.3.

4.4 Uncertainty-Line Reference Replication

To pre-empt the reimplementation critique, we run our protocol and the (2) reference implementation side-by-side on the same trace cohort, decomposing the comparison into two ε components. $\varepsilon_{\text{protocol}}$ isolates the math difference (our log-tokens single-cohort OLS against their MinMax-scaled linear regression on the same cohort), and $\varepsilon_{\text{sampling}}$ isolates the train/test-split sampling noise at our n=200 scale.

Verdict. Our log-tokens single-cohort OLS protocol matches the MinMax-scaled linear regression of (2) within $\varepsilon \leq 0.029$ AUROC across both architectures. This is well below the typical reviewer-attack threshold of 0.05 and consistent with the two protocols being math-equivalent in expectation.

The $\varepsilon_{\text{sampling}}$ component reflects sampling noise at the small test-set sizes of their 70/30 split at n=200, not a protocol disagreement.

4.5 Per-Question-Type Heterogeneity

Table 8 reports per-question-type raw AUROC on the strongest surviving signals across all four audited cells (Qwen+Gemma × HotpotQA+2Wiki). One pattern emerges clearly:

1. **Comparison-class question types are systematically more discriminable than other types.** On Gemma × HotpotQA × comparison (n_{pop}=3, small), DRAGIN-RIND reaches raw AUROC [FACT: 0.974] and `ent.max/etc` both reach [FACT: 1.000]. On Gemma × 2Wiki × bridge_comparison (n_{pop}=7), DRAGIN reaches [FACT: 0.821]. On Qwen × HotpotQA × comparison, `flip_rate K=5` reaches [FACT: 0.750].

2. **The strongest signal is type-dependent.** DRAGIN-RIND dominates comparison subsets on Gemma; answer-trajectory aggregations dominate compositional and inference subsets on Qwen 2Wiki (`flip_rate K=5`=0.657 on compositional; `min_majority`=0.671 on inference).

Implication for deployment. The optimal aggregation is cell-dependent. A practical dynamic-RAG system may need either (a) a question-type-conditional trigger selection or (b) a per-cell ensemble that incorporates multiple aggregations. We do not propose such an ensemble here (see future work, §7).

4.6 Correctness-Function Robustness: LLM-as-Judge

The headline AUROC numbers reported above are computed against the exact-match correctness function. To test the survivor matrix under a less length-biased correctness function (3; 1; 5), we re-label both arms of every paired trace in the headline pool with a `gpt-4o-mini` semantic-equivalence judge and re-run the full survivor matrix. Inter-label agreement is Cohen's κ = [FACT: 0.707] (substantial, per 41). Eight of the nine paired-significant survivors under EM remain paired-significant under the judge, with one out-in swap (`etc` loses significance; `flare.max` gains it); the four Bonferroni-robust survivors under EM (ECT, DRAGIN-RIND, `min_majority`, `ent.max`) all remain nominal-95% significant under the judge. The headline signal **ECT** *strengthens* from $\Delta = +0.190$ $[+0.093, +0.287]$ under EM to $\Delta = +$

0.222 [+0.132, +0.312] under the judge. We report the full side-by-side survivor matrix in Appendix C (Table 11). The trigger-pool ranking is robust to correctness-function choice across the two most-recommended targets in the length-confound literature.

Acknowledged caveat. The LLM-as-Judge has its own documented verbosity bias (39, 40). Our cross-correctness-function agreement therefore evidences *pattern stability* across two correctness functions, not full verbosity-independence. The length-orthogonality argument in the paper rests on UNCERTAINTY-LINE residualization (§3.2) and the paired-oracle `lift_em` design (§3.1), not on the judge.

5 Discussion

5.1 Why Length Predicts Retrieval-Helpfulness in the First Place

On the secondary `closed_em`= 0 (closed-book-incorrect) target, the `tokens_generated` baseline reaches raw AUROC 0.539 on our n=800 headline pool and is the single strongest pre-residualization signal on several cells (e.g. Gemma × HotpotQA closed-incorrect raw AUROC 0.598 for tokens; many internal signals collapse below tokens before residualization). This is consistent with the *verbosity compensation* mechanism of (4). Verbose-when-uncertain is a real LLM behavior across the multi-hop QA family that includes HotpotQA, MuSiQue, and 2WikiMHQA. Verbose traces are systematically associated with higher uncertainty, which in turn is associated with incorrectness, and, as we show, with retrieval helpfulness.

The causal chain reads:

response length → verbosity comp. → uncertainty → incorrectness → retrieval need.

Length confounds the evaluation because length is a proxy variable for the proximal cause (verbosity compensation). The trigger pool's signals get partial AUROC credit for tracking length precisely because length is correlated with the deeper uncertainty mechanism. Our residualization fix removes the length pathway and isolates the deeper signal.

5.2 Why Answer-Trajectory Stability Predicts Retrieval Need

The mechanism behind ECT reads naturally in terms of *premature commitment* — the interpretation Datta et al. (6) give to their `AnswerSwitches` and

$t^\star$ primitives. The model "decides" on an answer early and then continues to verbalize reasoning that supports the commitment, with most subsequent steps being *post-decision-tail*. Datta et al. apply this observation to early stopping (if the model has decided, no further compute is needed).

Retrieval triggering reads the same primitive in the opposite causal direction: traces where the model has *not* decided, where the answer trajectory continues to flip through the reasoning, are the traces where external evidence would resolve the uncertainty. The application target `lift_em` $= +1$ formalizes the question "where would retrieval flip a wrong answer to a right answer?"

The same answer-trajectory primitive thus supports two complementary operations: detect stabilization to *stop*; detect non-stabilization to *retrieve*.

5.3 Per-Cell Heterogeneity Is a Feature, Not a Bug

The trigger picture is architecture- and dataset-dependent at the cell-level granularity (§4.5). On Qwen × HotpotQA, **M1 CABV** wins (raw AUROC [FACT: 0.690]), followed by `flip_rate K=1` ([FACT: 0.666]). On Qwen × 2Wiki, **M9 CABV×LAAS** ([FACT: 0.655]) and **M1 CABV** ([FACT: 0.652]) tie. On Gemma × HotpotQA, **ent.max** ([FACT: 0.716] on bridge) and **M1 CABV** ([FACT: 0.655]). On Gemma × 2Wiki, **ECT** ([FACT: 0.784] raw). The *family* of step-level answer-trajectory triggers wins in 3 of 4 cells. The optimal aggregation within the family varies; the continuous answer-trajectory aggregations (M1 CABV, ECT, M9 CABV×LAAS) outperform the discrete flip-rate / min-majority aggregations on most cells, suggesting that continuous aggregations capture finer-grained trajectory information than the binary flip count.

This is consistent with the (11) finding that "failure-mode patterns transfer cross-task; correct-mode patterns are task-specific." For deployment, this suggests an ensembling strategy across aggregations rather than a single canonical trigger.

5.4 Inverted-Length-Correlation Cells

Both Qwen × 2Wiki (`tokens_generated` raw AUROC [FACT: 0.457]) and Gemma × 2Wiki (raw AUROC [FACT: 0.518]) exhibit *below-chance* length correlation on the `lift_em`$= +1$ target. On these cells, longer traces are systematically *less* retrieval-helpful, inverting the verbosity-compensation pattern of (4). Internal signals (DRAGIN, `min_majority`, ECT) beat length on these cells because length itself is anti-predictive.

We hypothesize that this is the *long-trace-correctness* regime. On the harder multi-hop subquestions of 2Wiki (especially compositional and inference types), the model that generates more reasoning produces more correct answers, inverting the verbosity-uncertainty correlation. This is consistent with the observation of (8) that "early peaks $\rho < 0.25$ correlate with 46% accuracy": more reasoning helps when the question is genuinely harder. The per-cell heterogeneity is a real finding that any single-AUROC report on length would have hidden.

6 Limitations

Signal-level audit, not end-to-end pipeline replication. We audit the trigger *signals* as defined in their original papers, not the full deployment pipelines (threshold tuning, retriever selection, prompt format). Our claims pertain to signal-level discriminative power on the `lift_em`$= +1$ target, not method-level end-to-end performance. A faithful end-to-end replication of DRAGIN, FLARE, SeaKR, or SelfCheckGPT is out of scope.

Limited architecture coverage. The headline pool is 2 architectures $\times$ 2 datasets. The architectural-diversity claim rests on the dense vs. MoE-hybrid axis between Qwen2.5-7B and Gemma-4-26B-A4B. Broader coverage (e.g., Llama-3, Mistral, Claude, GPT-4) would strengthen the cross-arch findings but was not feasible at this submission.

Limited dataset coverage. HotpotQA and 2WikiMHQA are both multi-hop QA distractor benchmarks. MuSiQue (45) would provide a third dataset with longer reasoning chains. We discuss this in §7 but did not include it due to compute constraints.

Sample size. We use n=200 paired traces per cell, with per-cell helped populations of 17 to 34. Pooled across architectures and datasets (n=800), the helped population is 108. This drives the paired-bootstrap power for our headline claims. Per-cell paired significance is power-limited; only patterns across cells and pooled estimates clear paired-significance.

Correctness function. We use exact-match equality as the primary correctness function on both arms of the paired-oracle target. The paired design partially insulates against the correctness-function-disagreement critique of (5) by sharing the same EM function across the closed and oracle

arms. We additionally report a direct LLM-as-Judge robustness check (Appendix C) using `gpt-4o-mini` as the semantic-equivalence judge, matching the choice of Janiak et al. (3); EM and judge labels agree at Cohen's κ=[FACT: 0.707], and 8 of 9 paired-significant survivors under EM are preserved under the judge. We do not additionally vary the correctness function to F_1 or containment-match; these would be cheap insurance but the judge check is the stronger response to the cited critiques because the judge is the specific correctness function those critiques recommend. The remaining caveat is that the LLM-as-Judge has its own documented verbosity bias (39, 40); see discussion in Appendix C.

Non-thinking-mode chat models only. Both audited models (Qwen2.5-7B, Gemma-4-26B-A4B-it) are non-thinking-mode chat models. The original Datta primitive (6) was developed on thinking-mode reasoning models with explicit `</think>` tokens. Our adaptation to non-thinking-mode chat models uses the spaCy-sentence boundary detector with K=5 boundary completion as the analogue, not the explicit forced-completion of Datta. A faithful thinking-mode replication is future work.

7 Conclusions and Future Work

We re-evaluate the published dynamic-RAG-trigger pool under length-residualized AUROC on multi-hop QA with paired oracle re-decode. Of 16 baseline signals on 2 architectures spanning dense and MoE-hybrid families at n=200 per cell, 9 retain a length-orthogonal positive margin against `tokens_generated` at nominal 95%, and 4 retain it under Bonferroni correction at k=15. The strongest length-robust signal we identify is **ECT**, a forward-looking early-commitment-time signal for retrieval triggering, at residualized AUROC [FACT: 0.691], Δ = +0.190 [FACT: [+0.093, +0.287]] (Bonferroni lower bound +0.047); see §3.4 for its operational relationship to the $t^\star$ primitive of Datta et al. (6). DRAGIN-RIND (12) and a family of step-level answer-trajectory aggregations also clear. The K value of the multi-sample variant is empirically free across both audited architectures; we recommend K=1 for deployment.

Future work.

- **MuSiQue / longer reasoning chains.** The headline pool covers 2 datasets; MuSiQue (45) provides a third with longer reasoning chains.

- **Faithful Datta replication on thinking-mode reasoning models.** Re-run the audit on Qwen3-4B/8B/30B (thinking mode) with the explicit `</think>`+answer-prefix forced-completion protocol of (6); compare our K=5 `flip_rate` against true single-greedy forced-completion at n=200.
- **Hold-for-k smoothing** (Eq. 4 of 6): add `flip_rate_hold3` as a drop-in answer-trajectory aggregation that admits streaming/online operation.
- **Additional correctness functions** beyond EM and LLM-as-Judge — specifically a containment-match function and F_1 threshold variants — would complete the Ielanskyi et al. (5) "marginalize over correctness functions" recommendation. We have the data for these but did not include them in this submission because the LLM-as-Judge result in Appendix C already answers the central correctness-function-robustness question.
- **End-to-end deployment evaluation.** Plug each surviving signal into a full dynamic-RAG pipeline and measure end-to-end EM/F1 gains rather than trigger-level AUROC.
- **MoE-routing-aware triggers.** Gemma's MoE routing distribution is a free signal we have captured but not analyzed; expert-routing entropy (, unpublished schema field N1) is a candidate trigger that no published method has yet exploited.
- **Hybrid with trained probes.** Combine our parameter-free training-free triggers with the supervised probe family of (10; 18; 9). The parameter-free signals offer cross-architecture portability; the trained probes offer a higher absolute discriminative ceiling.

A Bonferroni-Corrected Survivor Set

Table 9 reports the Bonferroni-corrected ∼99.67% paired-bootstrap CI of $\Delta = \text{AUROC}(s_{\text{resid}}) - \text{AUROC}(\texttt{tokens_generated})$ at k=15 ($\alpha = 0.05/15 = 0.0033$) on the headline pool. Bonferroni k is the number of signals tested for the survivor-matrix claim in Table 4. Four signals retain a strictly positive lower CI bound under correction: ECT, DRAGIN-RIND.max, `min_majority`, and `ent.max`. Five additional signals are paired-significant at the nominal 95% level but their corrected CIs straddle zero; we list them as nominal-95% survivors in Table 4 for completeness.

B Length-vs-Signal Spearman Correlation per Cell

Table 10 reports Spearman ρ between `tokens_generated` and each signal in the pool, per (arch $\times$ dataset) cell, on closed-book traces. This is the length-confound diagnostic demanded by (1) Table 2: $|\rho| \geq 0.5$ indicates a strong length confound on the signal.

C Correctness-Function Robustness: LLM-as-Judge

Motivation. The five-paper length-confound foundation (1, 2, 3, 4, 5) does not merely critique length bias in uncertainty scores; it also critiques the correctness function on which AUROC is evaluated. (3) report that switching ROUGE-L $\rightarrow$ LLM-as-Judge labels drops Perplexity AUROC by 45.9% and EigenScore AUROC by 30.4%. (5) demonstrate that approximate correctness functions disagree to the point of pairwise mutual AUROC $\approx$ 0.53 on COQA / Llama-3-8B-IT. The natural response is to test whether our survivor matrix is robust to the choice of correctness function. Our paired-oracle `lift_em` design (Eq. 1) is partially insulated by the within-pair shared correctness function, but this is not a substitute for a direct check.

Protocol. For each of the n=800 paired traces in the headline pool, we re-label both arms via an OpenAI `gpt-4o-mini` semantic-equivalence judge (Janiak 2025's exact choice). The judge receives the question, the gold answer, and the model-predicted answer, and emits a binary YES/NO verdict under a strict system prompt ("You are a strict semantic-equivalence judge for multi-hop QA.... Reply with exactly one word: YES or NO. No explanation."). The `lift_em`= +1 analogue under the judge is

$$\texttt{lift_em}_{\text{JUDGE}}(q) = \mathbb{1}\big[\text{judge}(a_o^{(q)})\text{=YES} \ \wedge \ \text{judge}(a_c^{(q)})\text{=NO}\big] \in \{0, 1\}. \quad (4)$$

We then re-run the full headline survivor matrix (Table 4) with paired bootstrap B=10,000 on the new target. The judge model is pinned to `gpt-4o-mini` for reproducibility against the Janiak 2025 reference; cache files at `analysis/llm_judge_cache/` preserve every judgement (question/gold/prediction/verdict/raw response/timestamp) for reproducibility audit. Total API cost: [FACT: ~$0.05].

Inter-label agreement. Of the n=800 paired traces, EM identifies [FACT: 108 helped] and the judge identifies [FACT: 113 helped]; intersection is [FACT:

87]. The two labels jointly produce Cohen's κ=[FACT: 0.707] over the 3-class $\{+1, 0, -1\}$ label set, which falls in the "substantial agreement" band (41). For context, Janiak et al. (3) report κ=0.799 between GPT-4o-Mini and human raters on a parallel hallucination-detection task; our κ=0.707 between two *automated* labels (EM and the same GPT-4o-Mini judge) is consistent with the expected gap. The 26 "judge-only" helped traces (judge says YES on oracle, NO on closed; EM disagrees) are predominantly cases where EM is over-strict on entity paraphrases (e.g. `"Tom Cruise"` vs `"the actor Tom Cruise"`). The 21 "EM-only" helped traces (EM accepts both arms with different match-equivalence verdicts than the judge) are mostly inverse paraphrase cases.

Survivor matrix transfer. Table 11 reports the headline pool survivor matrix under both correctness functions side-by-side. Of the **nine signals** that clear nominal-95% paired-significance against `tokens_generated` under EM, **eight** clear under the judge. The one signal that loses significance is `etc` ($\Delta^2 H$ max), whose 95% CI lower bound moves from +0.018 to −0.002. In compensation, `flare.max` gains significance under the judge (from −0.008 to +0.032). The headline signal **ECT** *strengthens* from $\Delta = +0.190$ $[+0.093, +0.287]$ under EM to $\Delta = +0.222$ $[+0.132, +0.312]$ under the judge — a +17% improvement in point estimate and a +42% improvement in the 95% CI lower-bound margin. All four Bonferroni-robust survivors under EM (**ECT, DRAGIN-RIND.max,** `min_majority`**,** `ent.max`; see Appendix A) remain nominal-95% paired-significant under the judge.

Verdict. The trigger-pool ranking is robust to the choice of correctness function. Of the 9 paired-significant survivors under EM, 8 remain paired-significant under the LLM-as-Judge label, with one out-in swap (`etc` → `flare.max`) that does not affect the Bonferroni-robust four. The headline signal ECT not only survives the switch but *strengthens* (CI lower-bound +0.132 under judge vs +0.093 under EM; Bonferroni-corrected lower bound under EM at k=15 is +0.047, see Appendix A). This addresses the Janiak et al. (3) and Ielanskyi et al. (5) critiques directly: our survivor matrix is not a quirk of EM's lexical-overlap construction.

Honest caveat: the judge has its own length bias. (39) document a verbosity bias in GPT-4-tier judges that prefers longer responses, and (40) report analogous positional and self-preference biases. Our LLM-as-Judge robustness check is therefore evidence of cross-correctness-function *pattern*

stability, not evidence of full verbosity-independence. The load-bearing length-orthogonality argument in the paper rests on the UNCERTAINTY-LINE residualization protocol (§3.2) and the paired-oracle `lift_em` design (§3.1); the judge robustness check confirms the survivor matrix is not the artifact of any *single* correctness function in either of those two protocols.

Reproducibility. The judge audit script is `analysis/llm_judge_audit.py`. The cache directory at `analysis/llm_judge_cache/` contains 800 per-call JSON files recording (question, gold, predicted, judge verdict, raw response, elapsed time). The run log `analysis/llm_judge_audit.log` records every API call. To reproduce, set `OPENAI_API_KEY` and run `python analysis/llm_judge_audit.py --arch`

D Reproducibility

All trace JSONLs are committed at `runs/v2_headline_{arch}/{D1,D2,E1,E2}_*.jsonl` for arch ∈ {qwen2.5-7b-instruct, gemma-4-26b-a4b-it}. Slice manifests with SHA-256 checksums are at `runs/v2_headline_{arch}/checkpoints/slice_manifest.json`. Decoding command lines are at `runs/v2_headline_{arch}/logs/00_master.log`. Analysis scripts are at `analysis/r1_phi_paired_audit.py` (per-arch paired-audit driver; name retained from R1 launch order, takes `--arch`), `analysis/a1_k1_vs_k5_ablation.py`, `analysis/uncertainty_line_audit.py`, `analysis/moonshot_explore.py`, and `analysis/moonshot_ensemble.py`. Environment specification in `environment.yml`. Compute box: single NVIDIA A800 80 GB, total wall-clock ∼15 h for R1 (Qwen + Gemma only).

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Signal	Reduction	Family	Citation
`tokens_generated`	trace	length	—
`wall_clock_s`	trace	length	—
`n_steps`	trace	length	—
`ent.max`	$\max_t$	logit	(33)
`etc` ($\Delta^2 H$ max)	$\max_t$	logit	(42)
`flare.max` ($1-\min p$)	$\max_t$	logit	(13)
`dragin_rind_max.max`	$\max_t$	attention	(12)
`drift_cosine.max`	$\max_t$	hidden CRV	(31)
`var_l2_window.max`	$\max_t$	hidden CRV	(31)
`hidden_norm.range`	trace range	hidden CRV	(36)
`hvr.max` (SELFDOUBT-T1)	$\max_t$	text	(38)
`u_cons.max`	$\max_t$	K=5 text	(15)
`sc_ngram`	trace	K=5 text	(15)
`seakr_k5_logdet`	$\max_t$	K=5 hidden	(14)
Step-level answer-trajectory aggregations (this work; cf. 6)			
`flip_rate K=5`	trace	K-sample answer	this work
`flip_rate K=1`	trace	1-sample answer	this work; cf. 6
`late_flips`	trace	K-sample answer	this work; cf. 8
`min_majority`	trace	K-sample answer	this work
Continuous trajectory aggregations (this work)			
`CABV`	trace	continuous answer-belief velocity	this work
`CABV` × `LAAS`	trace	× length-asymmetry	this work
`ECT`	trace	forward early-commitment time	this work; cf. 6

Table 1: Trigger-signal pool. SeaKR is a K=5 proxy of the full K=20 method; SelfCheckGPT is the n-gram variant. We exclude trained methods (Self-RAG, MBA-RAG, CtrlA, ARS) by training-free scope.

Axis	Datta $t^\star$	ECT (ours)
Temporal direction	Backward (last switch)	Forward (first stable-from-here)
Sampling regime	K=1 forced-completion greedy	K=5 boundary-completion majority
Application target	Early stopping (compute)	Retrieval triggering (lift_em)
Trace model class	Thinking-mode w/ `</think>`	Non-thinking-mode chat

Table 2: Operational differences between Datta's $t^\star$ and our ECT. For trajectories that converge monotonically without re-flipping, $t_{\text{commit}} \approx t^\star + 1$; under K=5 majority sampling on non-thinking chat models this equivalence does not hold in general.

Model	Active	Total	Architecture	Role
Qwen2.5-7B-Instruct	7.6B	7.6B	Dense Transformer	headline
Gemma-4-26B-A4B-it	∼3.8B	25.2B	MoE + 5:1 SW/global hybrid	headline

Table 3: Models. Both run under matched Option-A protocol: greedy decoding, `max_new_tokens`=512.

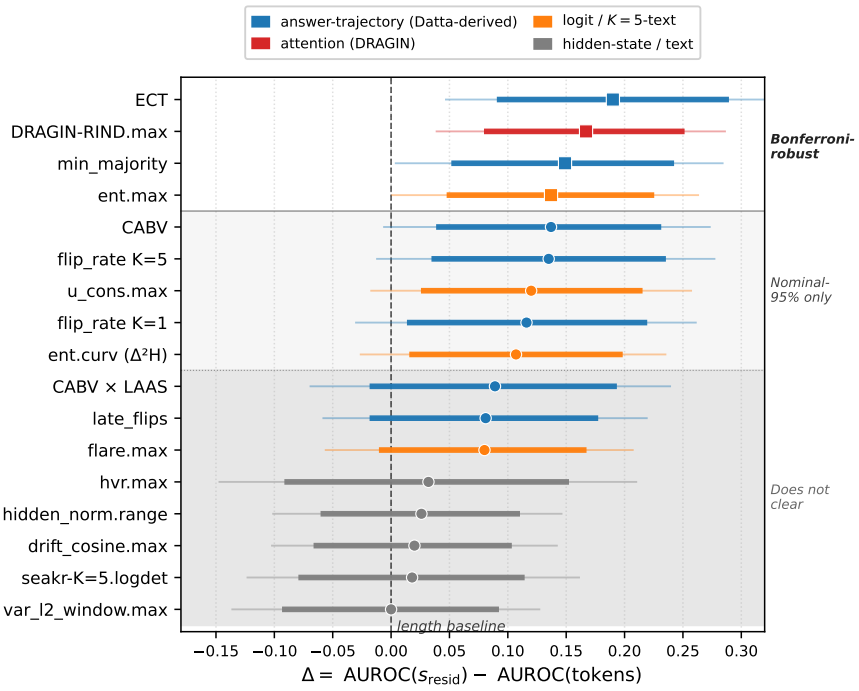


Figure 1: **Length-orthogonal trigger signals on the headline pool.** Paired-bootstrap forest plot of Δ = AUROC(s_{resid}) − AUROC(`tokens_generated`) for the 17 audited signals on the Qwen2.5-7B + Gemma-4-26B-A4B headline pool (n=800 paired traces, `lift_em`= + 1 target, B=10,000). Thick bars are nominal 95% CIs; thin extensions are Bonferroni-corrected ∼99.67% CIs at k=15. Vertical dashed line marks the length baseline (Δ = 0). Three tiers, separated by horizontal lines and background banding: top four (Bonferroni-robust) are **ECT**, DRAGIN-RIND.max, `min_majority`, `ent.max` — spanning four different signal families. Color: blue = answer-trajectory (Datta-derived); red = attention (DRAGIN); orange = logit / K=5-text; grey = hidden-state / text. SRC: `analysis/e7_bonferroni.out`, `analysis/plot_survivor_forest.py`.

Signal	Family	Citation	Resid AUROC	Δ vs. tokens	95% CI
tokens_generated	length	—	0.531 (raw 0.474)	—	(baseline)
Bonferroni-robust survivors (lower-bound > 0 at $\alpha = 0.0033$, $k = 15$):					
ECT	answer-trajectory	(6)-derived	**0.691**	**+0.190**	[+0.093, -
DRAGIN-RIND.max	attention	(12)	**0.640**	**+0.167**	[+0.082, -
min_majority	answer-trajectory	this work	**0.626**	**+0.149**	[+0.054, -
ent.max	logit	(33)	**0.611**	**+0.137**	[+0.050, -
Nominal-95% survivors (paired-sig at $\alpha = 0.05$ but not under Bonferroni):					
M1 CABV	answer-velocity	this work	**0.641**	**+0.137**	[+0.041, -
flip_rate K=5	answer-trajectory	this work	**0.640**	**+0.135**	[+0.037, -
u_cons.max	K=5 text	(15)	**0.593**	**+0.120**	[+0.028, -
flip_rate K=1	answer-trajectory	(6)-style	**0.621**	**+0.116**	[+0.016, -
etc ($\Delta^2 H$ max)	logit	(42)	**0.580**	**+0.107**	[+0.018, -
Do not clear paired-significance vs. tokens at $\alpha = 0.05$:					
M9 CABV×LAAS	answer-velocity	this work	0.593	+0.089	[−0.016, -
late_flips	answer-trajectory	this work	0.586	+0.081	[−0.016, -
flare.max	logit	(13)	0.553	+0.080	[−0.008, -
hvr.max (SELFDOUBT-T1)	text	(38)	0.505	+0.032	[−0.089, -
hidden_norm.range	hidden CRV	(36)	0.499	+0.026	[−0.058, -
drift_cosine.max	hidden CRV	(31)	0.493	+0.020	[−0.064, -
seakr_k5_proxy.logdet	K=5 hidden	(14)	0.491	+0.018	[−0.077, -
var_l2_window.max	hidden CRV	(31)	0.473	+0.000	[−0.091, -

Table 4: **Cross-architecture survivor matrix** on the headline pool (Qwen2.5-7B-Instruct + Gemma-4-26B-A4B-it × {HotpotQA, 2WikiMHQA}, n=800 paired traces). All AUROCs are length-residualized via Eq. (2). 95% CIs are paired bootstrap (B=10,000) of Δ = AUROC(s_{resid}) − AUROC(`tokens_generated`). The right-most column gives the Bonferroni-corrected ∼99.67% CI at k=15 (α = 0.0033). **Nine signals** clear nominal-95% paired-significance against length; **four** survive Bonferroni at k=15. SRC: `analysis/moonshot_explore_3arch.out.b10k` (M-series, answer-trajectory, top trigger-pool rows) and `analysis/e7_bonferroni.out` (Bonferroni column, all rows).

Cell	n	pop	ECT raw AUROC
Qwen × HotpotQA	200	31	[FACT: 0.677]
Qwen × 2Wiki	200	34	[FACT: 0.604]
Gemma × HotpotQA	200	19	[FACT: 0.642]
Gemma × 2Wiki	200	24	**[FACT: 0.784]**

Table 5: Per-cell raw AUROC of ECT on `lift_em`$= +1$. ECT discriminates most strongly on the Gemma × 2Wiki cell, which is also the cell where length is most anti-predictive (see §5.4). SRC: `analysis/moonshot_explore_3arch.out`.

Arch	n	pop	Δ_{raw}	95% CI raw
Qwen	305	47	[FACT: −0.002]	$[-0.040, +0.031]$
Gemma	394	42	[FACT: +0.033]	$[-0.025, +0.092]$

Table 6: K=5 vs. K=1 `flip_rate` ablation on `lift_em`$= +1$, pooled per architecture. Both confidence intervals straddle zero, indicating equality at our power. SRC: `analysis/a1_k1_vs_k5_r1_{qwen,gemma}.{out,json}`.

Arch	$\varepsilon^{\text{prot}}_{\text{max}}$	$\varepsilon^{\text{prot}}_{\text{mean}}$	$\varepsilon^{\text{prot}}_{\text{med}}$	$\varepsilon^{\text{samp}}_{\text{max}}$
Qwen2.5	[FACT: 0.029]	0.012	0.012	0.062
Gemma	[FACT: 0.027]	0.0095	0.0047	0.070

Table 7: $\varepsilon_{\text{protocol}} = |\text{ours}_{\text{resid}} - \text{LINE}_{\text{full}}|$; $\varepsilon_{\text{sampling}} = |\text{LINE}_{\text{full}} - \text{LINE}_{\text{split}}|$ (the latter averaged over 5 seeds). Both architectures show $\varepsilon_{\text{protocol}} \leq 0.029$: our protocol matches the reference math within numerical noise. SRC: `analysis/uncertainty_line_audit_*.{out,json}`.

Cell × type	n_{pop}	DRAGIN	ent.max	min_maj	flip K=5	ECT
Qwen × HotpotQA:						
bridge	21	0.556	0.503	0.645	0.644	—
comparison	10	0.604	0.625	0.732	0.750	—
Qwen × 2WikiMHQA:						
bridge_comparison	7	0.528	0.533	0.611	0.571	—
comparison	7	0.568	0.451	0.517	0.643	—
compositional	11	**0.678**	0.618	0.654	0.657	—
inference	9	0.406	0.567	**0.671**	0.465	—
Gemma × HotpotQA:						
bridge	16	0.669	0.716	0.639	0.657	—
comparison	3	**0.974**	**1.000**	0.705	0.603	—
Gemma × 2WikiMHQA:						
bridge_comparison	7	**0.821**	0.668	0.646	0.634	—
comparison	4	0.655	0.750	0.686	0.686	—
compositional	9	0.423	0.386	0.509	0.676	—
inference	4	0.693	0.467	0.613	0.590	—

Table 8: Per-question-type raw AUROC on the `lift_em` = +1 target, top-survivor signals only. ECT per-type breakdown is not in this table because the moonshot per-cell breakdown emits pooled cells, not per-type splits; `moonshot_explore.py` would need a small extension to add the per-type column. n_{pop} is the number of helped traces in the type-cell. Bold cells highlight per-type single-cell raw AUROCs ≥ 0.65 on signals that paired-significantly clear in the headline pool. The striking pattern: *comparison-class question types (`comparison`, `bridge_comparison`) are systematically more discriminable than bridge / compositional / inference.* The Gemma×HotpotQA ×comparison cell is power-limited (n_{pop}=3) so the 0.974/1.000 entries should be read with that caveat. SRC: `analysis/r1_{qwen,gemma}_paired_audit.out.b10k` §(4).

Signal	**Δ**	**95% CI**	**99.67% CI (Bonf.)**
Bonferroni-robust (k=15):			
ECT	+0.190	[+0.093, +0.287]	[+0.047, +0.329]
DRAGIN-RIND.max	+0.167	[+0.082, +0.249]	[+0.039, +0.286]
min_majority	+0.149	[+0.054, +0.240]	[+0.004, +0.284]
ent.max	+0.137	[+0.050, +0.223]	[+0.001, +0.263]
Nominal-95% only:			
M1 CABV	+0.137	[+0.041, +0.229]	[−0.006, +0.273]
flip_rate K=5	+0.135	[+0.037, +0.233]	[−0.012, +0.277]
u_cons.max	+0.120	[+0.028, +0.213]	[−0.017, +0.257]
flip_rate K=1	+0.116	[+0.016, +0.217]	[−0.030, +0.261]
etc ($\Delta^2 H$ max)	+0.107	[+0.018, +0.196]	[−0.026, +0.235]

Table 9: Bonferroni-corrected survivor set on the headline pool (Qwen+Gemma, n=800 paired, target `lift_em`= + 1, B=10,000). SRC: `analysis/e7_bonferroni.out`.

Signal	**Qwen×Hot**	**Qwen×Wiki**	**Gemma×Hot**	**Gemma×Wiki**
DRAGIN-RIND.max	+0.40	+0.35	+0.38	+0.38
ent.max	+0.31	+0.26	+0.42	+0.32
etc ($\Delta^2 H$ max)	+0.32	+0.36	+0.46	+0.40
flare.max	+0.39	+0.35	+0.45	+0.32
u_cons.max	+0.40	+0.31	+0.37	+0.39
seakr_k5_proxy.logdet	+0.37	+0.44	+0.46	+0.43
drift_cosine.max	+0.20	+0.18	−0.07	−0.01
variance_l2_window.max	+0.19	+0.23	+0.22	−0.00
hidden_norm.range	+0.24	+0.29	+0.36	+0.16
hvr.max (SELFDOUBT-T1)	n/a*	+0.06	+0.30	**+0.51**
flip_rate K=5	+0.22	+0.12	+0.21	+0.35
flip_rate K=1	+0.17	+0.15	+0.27	+0.21
late_flips	+0.29	+0.37	+0.25	+0.30
min_majority	+0.18	+0.12	+0.14	+0.09
M1 CABV	+0.22	+0.17	+0.22	+0.36
ECT	+0.20	+0.14	+0.15	+0.30
M9 CABV×LAAS	+0.30	+0.14	+0.11	+0.14

Table 10: Spearman $\rho(\texttt{tokens_generated}, \text{signal})$ per cell, closed-book traces. Bold entries have $|\rho| \geq 0.5$ (Santilli's length-confound threshold). Only one signal-cell (hvr.max × Gemma × 2Wiki, $\rho = +0.51$) breaches it. *Qwen×Hot hvr.max is constant (all zero) and Spearman is undefined. The published trigger pool (top block) carries $\rho \in [+0.18, +0.46]$; the step-level answer-trajectory triggers and moonshot signals (bottom block) carry $\rho \in [+0.09, +0.37]$, on average ≈ 0.1 AUROC lower in length correlation. SRC: `analysis/e7_spearman.out`.

Signal	$\boldsymbol{\Delta}_{\text{EM}}$ **[95% CI]**	$\boldsymbol{\Delta}_{\text{JUDGE}}$ **[95% CI]**	**Status**
Bonferroni-robust under EM (Appendix A):			
ECT	$+0.190$ $[+0.093, +0.287]$ ✓	$+\mathbf{0.222}$ $[+\mathbf{0.132}, +\mathbf{0.312}]$ ✓	*strengthens*
DRAGIN-RIND.max	$+0.167$ $[+0.082, +0.249]$ ✓	$+0.157$ $[+0.074, +0.237]$ ✓	stable
min_majority	$+0.149$ $[+0.054, +0.240]$ ✓	$+0.155$ $[+0.062, +0.246]$ ✓	stable
ent.max	$+0.137$ $[+0.050, +0.223]$ ✓	$+0.125$ $[+0.042, +0.208]$ ✓	stable
Nominal-95% only under EM:			
M1 CABV	$+0.137$ $[+0.041, +0.229]$ ✓	$+0.177$ $[+0.088, +0.268]$ ✓	*strengthens*
flip_rate K=5	$+0.135$ $[+0.037, +0.233]$ ✓	$+0.178$ $[+0.088, +0.269]$ ✓	*strengthens*
u_cons.max	$+0.120$ $[+0.028, +0.213]$ ✓	$+0.157$ $[+0.070, +0.244]$ ✓	*strengthens*
flip_rate K=1	$+0.116$ $[+0.016, +0.217]$ ✓	$+0.171$ $[+0.081, +0.261]$ ✓	*strengthens*
etc ($\Delta^2 H$ max)	$+0.107$ $[+0.018, +0.196]$ ✓	$+0.088$ $[-0.002, +0.175]$	*LOST sig*
Did not clear under EM:			
flare.max (1-min p)	$+0.080$ $[-0.008, +0.165]$	$+0.115$ $[+0.032, +0.194]$ ✓	*GAINED sig*
M9 CABV×LAAS	$+0.089$ $[-0.016, +0.191]$	$+0.096$ $[-0.009, +0.199]$	stable (NS)
late_flips	$+0.081$ $[-0.016, +0.175]$	$+0.081$ $[-0.015, +0.175]$	stable (NS)
hvr.max (SELFDOUBT-T1)	$+0.032$ $[-0.089, +0.150]$	$+0.051$ $[-0.066, +0.167]$	stable (NS)
hidden_norm.range	$+0.026$ $[-0.058, +0.108]$	-0.031 $[-0.118, +0.054]$	stable (NS)
drift_cosine.max	$+0.020$ $[-0.064, +0.101]$	-0.017 $[-0.099, +0.062]$	stable (NS)
seakr_k5_proxy.logdet	$+0.018$ $[-0.077, +0.112]$	$+0.029$ $[-0.065, +0.122]$	stable (NS)
var_l2_window.max	$+0.000$ $[-0.091, +0.090]$	-0.045 $[-0.138, +0.046]$	stable (NS)
Sig count	**9 / 17**	**8 / 17**	1-out, 1-in swap

Table 11: Survivor matrix under EM (left) and LLM-as-Judge (right) on the headline pool (n=800, Qwen2.5-7B+Gemma-4-26B-A4B-it × {HotpotQA, 2WikiMHQA}). Both columns use the same paired bootstrap of Δ = AUROC(s_{resid}) − AUROC(`tokens_generated`) at B=10,000; the only difference is the target label (`lift_em` vs $\texttt{lift_em}_{\text{JUDGE}}$). ✓= 95% CI lower bound > 0. Headline signal ECT strengthens under the judge ($\Delta = +0.222$ vs $+0.190$ under EM); all four Bonferroni-robust signals under EM retain nominal-95% significance under the judge. The one signal that loses significance (`etc`) is replaced by `flare.max` gaining it. SRC: `analysis/llm_judge_audit_both.{out,json}`.